

Graphs with clique number four whose chromatic number exceeds the cochromatic number by exactly three: an affirmative answer to Problem 1.5 of Steiner

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Machine-generated note

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Abstract

The cochromatic number $\zeta(G)$ of a graph G is the least number of parts in a partition of $V(G)$ into cliques and independent sets. Steiner (arXiv:2408.02400) disproved a 1988 conjecture of Erdős, Gimbel and Straight by exhibiting infinitely many graphs with $\omega(G) < 5$, $\zeta(G) = 4$ and $\chi(G) = 7$, and asked (Problem 1.5) whether for every $k \geq 5$ there is a graph G_k with $\omega(G_k) < 5$, $\zeta(G_k) = k$ and $\chi(G_k) = k + 3$. We answer this affirmatively. The complement G of the circulant graph $C_{13}(1, 5)$ has $(\omega, \zeta, \chi) = (4, 4, 7)$ and admits an optimal cochromatic partition with an independent part; a lemma shows that the Mycielskian raises the cochromatic number by exactly one under this hypothesis and preserves the hypothesis, while it is classical that it raises the chromatic number by one and preserves the clique number. Hence the iterated Mycielskians $G_k = \mu^{k-4}(G)$, of order $14 \cdot 2^{k-4} - 1$, satisfy $\omega(G_k) = 4$, $\zeta(G_k) = k$ and $\chi(G_k) = k + 3$ for every $k \geq 4$.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. A machine-checked Rocq/MathComp formalization of this result is available as [cochromatic_gap_three_proved](#). [Section 6](#) records the full provenance.

1 Introduction

All graphs are finite and simple. A set of vertices is *homogeneous* if it is a clique or an independent set. A *cochromatic partition* of G is a partition of $V(G)$ into homogeneous sets, and the *cochromatic number* $\zeta(G)$ is the least number of parts of such a partition. We write ω, α, χ for the clique, independence and chromatic numbers. Clearly $\zeta(G) \leq \chi(G)$.

Erdős, Gimbel and Straight conjectured in 1988 that every graph G with $\omega(G) < 5$ and $\zeta(G) > 3$ satisfies $\chi(G) \leq \zeta(G) + 2$ (Conjecture 1.2 in [2]). Steiner [2] disproved this: his Theorem 1.4 gives infinitely many graphs G with $\omega(G) < 5$, $\zeta(G) = 4$ and $\chi(G) = 7$. He then asked whether the gap of three persists for every value of the cochromatic number.

Problem 1.1 ([2, Problem 1.5]). Let $k \geq 5$. Is there a graph G_k with $\omega(G_k) < 5$, $\zeta(G_k) = k$ and $\chi(G_k) = \zeta(G_k) + 3$?

Theorem 1.2 (Main theorem). *For every integer $k \geq 4$ there is a graph G_k on $14 \cdot 2^{k-4} - 1$ vertices with*

$$\omega(G_k) = 4, \quad \zeta(G_k) = k, \quad \chi(G_k) = k + 3.$$

Explicitly, G_4 is the complement of the circulant graph $C_{13}(1, 5)$ and $G_{k+1} = \mu(G_k)$ is the Mycielskian of G_k .

Corollary 1.3. *The answer to Problem 1.1 is affirmative for every $k \geq 5$.*

Proof. Take G_k from Theorem 1.2: $\omega(G_k) = 4 < 5$, $\zeta(G_k) = k$ and $\chi(G_k) = k + 3 = \zeta(G_k) + 3$. $\square$

Idea of proof. Two ingredients. (i) A 13-vertex *seed*: the complement G of the circulant graph $F = C_{13}(1, 5)$. The graph F is triangle-free and has independence number 4, so $\omega(G) = 4$ and $\alpha(G) = 2$; hence $\chi(G) \geq \lceil 13/2 \rceil = 7$ and $\zeta(G) \geq \lceil 13/4 \rceil = 4$, and explicit partitions show both bounds are attained, the optimal cochromatic partition having an independent part $\{8, 9\}$. (ii) A *cochromatic Mycielski lemma*: $\zeta(\mu(H)) \geq \zeta(H) + 1$ always, with equality whenever H has an optimal cochromatic partition containing an independent part, and this property is inherited by $\mu(H)$. Together with the classical facts $\chi(\mu(H)) = \chi(H) + 1$ and $\omega(\mu(H)) = \max\{\omega(H), 2\}$, iterating the Mycielskian from the seed raises ζ and χ in lockstep and keeps $\omega = 4$. The proof is carried out in Section 4.

2 The Mycielskian and the cochromatic number

For a graph H , the Mycielskian $\mu(H)$ has vertex set $V^0 \cup V^1 \cup \{r\}$, where $V^0 = \{v^0 : v \in V(H)\}$ (the *old layer*), $V^1 = \{v^1 : v \in V(H)\}$ (the *first layer*) and r is the *root*; its edges are

- u^0v^0 for every $uv \in E(H)$;
- u^0v^1 and u^1v^0 for every $uv \in E(H)$;
- rv^1 for every $v \in V(H)$;

and no others. Thus V^0 induces a copy of H , the first layer V^1 is independent, r is adjacent to all of V^1 and to nothing in V^0 , and v^0 is never adjacent to v^1 . For $S \subseteq V(H)$ we write $S^0 = \{v^0 : v \in S\}$ and $S^1 = \{v^1 : v \in S\}$.

Lemma 2.1 (Cochromatic Mycielski lemma). *For every graph H ,*

$$\zeta(\mu(H)) \geq \zeta(H) + 1.$$

If H has a minimum cochromatic partition containing an independent part, then $\zeta(\mu(H)) = \zeta(H) + 1$, and $\mu(H)$ again has a minimum cochromatic partition containing an independent part.

Idea of proof. Let $\mathcal{P}$ be a cochromatic partition of $\mu(H)$ into q parts and C the part containing r . If C is a clique it lies in $\{r\} \cup V^1$ and misses V^0 , so the other $q - 1$ parts restricted to $V^0 \cong H$ form a cochromatic partition of H . If C is independent, then $C = \{r\} \cup S^0$ for an independent $S \subseteq V(H)$; for each other part D let D^* consist of the v with $v^0 \in D$ together with the $v \in S$ with $v^1 \in D$. These $q - 1$ sets partition $V(H)$ and each is homogeneous, because a first-layer vertex v^1 has the same old-layer neighbours as v^0 , a clique D contains at most one first-layer vertex, and two substituted vertices of S are nonadjacent. Either way $\zeta(H) \leq q - 1$. For equality, lift a minimum partition $I, P_2, \dots, P_{\zeta(H)}$ of H with I independent to $\{r\} \cup I^0, P_2^0, \dots, P_{\zeta(H)}^0, V^1$; both $\{r\} \cup I^0$ and V^1 are independent. [\[full proof\]](#)

Lemma 2.2 (Mycielski [1]). *For every graph H , $\chi(\mu(H)) = \chi(H) + 1$ and $\omega(\mu(H)) = \max\{\omega(H), 2\}$.*

Idea of proof. These are the classical properties of the Mycielskian. Upper bound on χ : give v^0, v^1 the colour of v and r a new colour. Lower bound: in a proper q -colouring of $\mu(H)$ with r coloured q , no first-layer vertex has colour q ; recolour each v whose v^0 has colour q by the colour of v^1 . A clique of $\mu(H)$ through r has at most two vertices, and a clique avoiding r contains at most one first-layer vertex v^1 and not v^0 , so replacing v^1 by v^0 gives a clique of H of the same size. [\[full proof\]](#)

3 The seed

Let F be the circulant graph on $\mathbb{Z}_{13}$ with

$$xy \in E(F) \iff x - y \equiv \pm 1 \text{ or } \pm 5 \pmod{13},$$

i.e. $F = C_{13}(1, 5)$, a 4-regular graph on 13 vertices, and let $G = \overline{F}$ be its complement. For $x, y \in \mathbb{Z}_{13}$ the *cyclic distance* is the element of $\{0, 1, \dots, 6\}$ congruent to $\pm(x - y)$; two distinct vertices are adjacent in F exactly when their cyclic distance is 1 or 5, and adjacent in G exactly when it is 2, 3, 4 or 6.

Proposition 3.1 (The seed). *The graph $G = \overline{C_{13}(1, 5)}$ satisfies*

$$\omega(G) = 4, \quad \alpha(G) = 2, \quad \chi(G) = 7, \quad \zeta(G) = 4,$$

and the partition

$$A = \{0, 2, 4, 6\}, \quad B = \{1, 5, 7, 11\}, \quad C = \{3, 10, 12\}, \quad I = \{8, 9\}$$

of $\mathbb{Z}_{13}$ is a cochromatic partition of G into four parts in which A, B, C are cliques and I is independent. A proper 7-colouring of G is given by the classes $\{0, 1\}, \{2, 3\}, \{4, 5\}, \{6, 7\}, \{8, 9\}, \{10, 11\}, \{12\}$.

Idea of proof. F is triangle-free because no two distinct elements of $\{\pm 1, \pm 5\}$ differ by an element of $\{\pm 1, \pm 5\}$; and $\alpha(F) = 4$: $\{0, 2, 4, 6\}$ is independent, while five independent vertices would have five cyclic gaps, each at least 2, summing to 13, and in each of the three possible gap patterns two chosen vertices are at cyclic distance 5. Complementation gives $\omega(G) = 4$, $\alpha(G) = 2$, whence $\chi(G) \geq \lceil 13/2 \rceil = 7$ and $\zeta(G) \geq \lceil 13/4 \rceil = 4$; the displayed colouring and partition attain these bounds. [\[full proof\]](#)

4 Proof of the main theorem

Proof of Theorem 1.2. Let $G_4 = G = \overline{C_{13}(1, 5)}$ and $G_{k+1} = \mu(G_k)$ for $k \geq 4$, so that $G_k = \mu^{k-4}(G)$. Since $|V(\mu(H))| = 2|V(H)| + 1$ and $|V(G_4)| = 13 = 14 \cdot 2^0 - 1$, induction gives $|V(G_k)| = 14 \cdot 2^{k-4} - 1$.

We prove by induction on $k \geq 4$ that $\omega(G_k) = 4$, $\zeta(G_k) = k$, $\chi(G_k) = k + 3$, and that G_k has a minimum cochromatic partition containing an independent part. For $k = 4$ this is [Proposition 3.1](#). Assume it for k . By [Lemma 2.1](#), applied to $H = G_k$ (which has a minimum cochromatic partition with an independent part), $\zeta(G_{k+1}) = \zeta(G_k) + 1 = k + 1$ and G_{k+1} again has a minimum cochromatic partition with an independent part. By [Lemma 2.2](#), $\chi(G_{k+1}) = \chi(G_k) + 1 = k + 4$ and $\omega(G_{k+1}) = \max\{\omega(G_k), 2\} = 4$. This completes the induction. $\square$

5 Remarks

Remark 5.1 (The case $k = 4$). The seed itself has $(\omega, \zeta, \chi) = (4, 4, 7)$. This is not new: Steiner's Theorem 1.4 [2] already provides infinitely many such graphs (via an 11-vertex gadget of two joined 5-cycles rather than a circulant). The new content is the propagation to every $k \geq 5$, which is what [Problem 1.1](#) asks for.

Remark 5.2 (Where the independent part is needed). The inequality $\zeta(\mu(H)) \geq \zeta(H) + 1$ of [Lemma 2.1](#) holds for every H , but equality uses the existence of a minimum cochromatic partition with an independent part. Without it one would only obtain $\zeta(G_k) \geq k$, which does not pin down $\chi(G_k) - \zeta(G_k)$. The hypothesis holds for the seed (the part $I = \{8, 9\}$) and is shown to persist under μ , so no gap arises.

Remark 5.3 (Computational checks). The referee ([Appendix B](#)) verified by exhaustive computation that $\omega(F) = 2$, $\alpha(F) = 4$, $\omega(G) = 4$, $\alpha(G) = 2$, $\chi(G) = 7$ and $\zeta(G) = 4$, and that the displayed colouring and cochromatic partition are valid; that both assertions of [Lemma 2.1](#) hold for all 1099 labelled graphs on at most 5 vertices and for 95 random graphs on 6–8 vertices; and, independently of [Lemma 2.1](#), that $\mu(G)$ on 27 vertices has $\omega = 4$, $\zeta = 5$, $\chi = 8$. For $\mu^2(G)$ on 55 vertices it confirmed $\omega = 4$ directly; the exact values $\zeta = 6$, $\chi = 9$ there rest on the lemmas. Scripts: [verification/scripts/2408.02400__00/](#).

Remark 5.4 (Novelty). The referee’s search found only two papers citing [2], both on the Erdős–Gimbel random-graph question, and no resolution of [Problem 1.1](#); the catalog review of May 2026 found none either. This has not been checked by a human.

6 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The construction and proof were found by the language model gpt-5.6-so1 in a single autonomous pass on 2026-08-31, as part of a large sweep over open problems catalogued from arXiv (catalog id 2408.02400__00; original writeup in attacks/2408.02400__00/output.md). (2) An independent adversarial referee agent (claude-fable-5, 2026-09-02/03) re-derived every step, checked Problem 1.5 and Theorem 1.4 verbatim against the source paper, verified all parameters of the 13-vertex seed by exhaustive computation, tested the Mycielskian lemma on all 1099 labelled graphs with at most 5 vertices and on random larger graphs, and verified the $k = 5$ instance (27 vertices) end-to-end by brute force; its verdict was CONFIRMED. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by claude-fable-5.1: the text was reorganised, with statements and proof ideas in the main text and full proofs in [Appendix A](#); the standard upper bound $\chi(\mu(H)) \leq \chi(H) + 1$, which the writeup invoked as a “standard identity” without proof, is now spelled out; and context from the referee’s report was added as remarks. No mathematical content was changed. **No human mathematician has checked this argument.** We circulate it to obtain exactly that: is the proof correct, and is the result new?

(4) Independently of this note, the result has been formalized in Rocq/MathComp in the companion repository [LLM4Rocq/graph-theory-rocq](#) as [cochromatic_gap_three_proved](#), checked against a source-verified formal statement with no added axioms. That check certifies the formal proposition, not its faithfulness to the source paper.

A Full proofs

Lemma 2.1 (Cochromatic Mycielski lemma). *For every graph H ,*

$$\zeta(\mu(H)) \geq \zeta(H) + 1.$$

If H has a minimum cochromatic partition containing an independent part, then $\zeta(\mu(H)) = \zeta(H) + 1$, and $\mu(H)$ again has a minimum cochromatic partition containing an independent part.

Proof. Let $\mathcal{P}$ be a cochromatic partition of $\mu(H)$ into q parts, and let $C \in \mathcal{P}$ be the part containing the root r . We show $\zeta(H) \leq q - 1$; since $\mathcal{P}$ was arbitrary, this gives $\zeta(\mu(H)) \geq \zeta(H) + 1$.

Case 1: C is a clique. Every vertex of $C \setminus \{r\}$ is adjacent to r , so $C \subseteq \{r\} \cup V^1$; and since V^1 is independent, C contains at most one first-layer vertex and no old-layer vertex. Hence the $q - 1$ parts of $\mathcal{P} \setminus \{C\}$ cover V^0 . The restriction of a clique or an independent set to V^0 is a clique or an independent set of the induced subgraph $\mu(H)[V^0] \cong H$. Discarding empty restrictions, we obtain a cochromatic partition of H into at most $q - 1$ parts, so $\zeta(H) \leq q - 1$.

Case 2: C is independent. Since r is adjacent to every vertex of V^1 , $C \cap V^1 = \emptyset$, so $C = \{r\} \cup S^0$ for some $S \subseteq V(H)$; as S^0 is independent and V^0 induces H , the set S is independent in H . For every part $D \in \mathcal{P} \setminus \{C\}$ define

$$D^* = \{v \in V(H) : v^0 \in D\} \cup \{v \in S : v^1 \in D\}.$$

The sets D^ partition $V(H)$.* Let $v \in V(H)$. If $v \notin S$, then $v^0 \notin C$, so v^0 lies in exactly one part $D \neq C$, and v belongs to D^* through its first clause; v cannot belong to any D'^* through the second clause because $v \notin S$. If $v \in S$, then $v^0 \in C$, so v is in no D^* through the first clause, while $v^1 \notin C$ lies in exactly one part $D \neq C$, and $v \in D^*$ through the second clause. Thus each v lies in exactly one D^* .

Each D^* is homogeneous. Let $D \in \mathcal{P} \setminus \{C\}$ and let $u, v \in D^*$ be distinct. Note first that if $v \in S$ with $v^1 \in D$ then $v^0 \notin D$ (because $v^0 \in C$), so each $v \in D^*$ enters through exactly one of the two clauses, and we call v *old* if $v^0 \in D$ and *substituted* if $v \in S$ and $v^1 \in D$.

- If D is independent: two old vertices u, v have $u^0, v^0 \in D$ nonadjacent, so $uv \notin E(H)$. An old u and a substituted v have $u^0, v^1 \in D$ nonadjacent; since $u^0v^1 \in E(\mu(H))$ iff $uv \in E(H)$, we get $uv \notin E(H)$. Two substituted vertices lie in S , which is independent. So D^* is independent in H .
- If D is a clique: $r \notin D$, and D contains at most one first-layer vertex (as V^1 is independent), hence at most one substituted vertex. Two old vertices u, v have $u^0v^0 \in E(\mu(H))$, so $uv \in E(H)$. An old u and the substituted v have $u^0v^1 \in E(\mu(H))$, so $uv \in E(H)$ (and $u \neq v$ because $v^0 \notin D$). So D^* is a clique in H .

Discarding empty sets, $\{D^* : D \in \mathcal{P} \setminus \{C\}\}$ is a cochromatic partition of H into at most $q - 1$ parts, so $\zeta(H) \leq q - 1$.

Equality and persistence. Suppose $I, P_2, \dots, P_{\zeta(H)}$ is a minimum cochromatic partition of H with I independent. Consider

$$\{r\} \cup I^0, \quad P_2^0, \dots, P_{\zeta(H)}^0, \quad V^1.$$

These sets partition $V(\mu(H))$. The set $\{r\} \cup I^0$ is independent, since I^0 is independent and r has no neighbour in V^0 ; each P_i^0 is a clique or independent set because V^0 induces H ; and V^1 is independent. So this is a cochromatic partition of $\mu(H)$ into $\zeta(H) + 1$ parts, whence $\zeta(\mu(H)) \leq \zeta(H) + 1$. Combined with the lower bound, $\zeta(\mu(H)) = \zeta(H) + 1$, the displayed partition is minimum, and it contains the independent parts $\{r\} \cup I^0$ and V^1 . $\square$

Lemma 2.2 (Mycielski [1]). *For every graph H , $\chi(\mu(H)) = \chi(H) + 1$ and $\omega(\mu(H)) = \max\{\omega(H), 2\}$.*

Proof. Upper bound on χ . Given a proper $\chi(H)$ -colouring c of H , colour v^0 and v^1 with $c(v)$ and give r a new colour. Edges u^0v^0 and u^0v^1 join vertices with colours $c(u) \neq c(v)$, and edges rv^1 join the new colour to an old one. So $\chi(\mu(H)) \leq \chi(H) + 1$.

Lower bound on χ . Let c be a proper colouring of $\mu(H)$ with colours $1, \dots, q$, renamed so that $c(r) = q$. No first-layer vertex has colour q , since all of V^1 is adjacent to r . Define c' on $V(H)$ by $c'(v) = c(v^0)$ if $c(v^0) \neq q$ and $c'(v) = c(v^1)$ if $c(v^0) = q$; then c' uses only colours $1, \dots, q - 1$. Let $uv \in E(H)$. If neither endpoint is recoloured, $c'(u) = c(u^0) \neq c(v^0) = c'(v)$ because u^0v^0 is an edge. If exactly one is recoloured, say $c(v^0) = q$, then $c'(v) = c(v^1) \neq c(u^0) = c'(u)$ because u^0v^1 is an edge. Both cannot be recoloured, since u^0, v^0 are adjacent and cannot both have colour q . Hence c' is a proper $(q - 1)$ -colouring of H and $\chi(H) \leq \chi(\mu(H)) - 1$.

Clique number. A clique of $\mu(H)$ containing r lies in $\{r\} \cup V^1$ and, V^1 being independent, has at most two vertices. Let K be a clique avoiding r . It contains at most one first-layer vertex, say v^1 (if any), and then $v^0 \notin K$ since $v^0v^1 \notin E(\mu(H))$. Replacing v^1 by v^0 yields a set $K' \subseteq V^0$ of the same size, which is a clique because $u^0v^1 \in E(\mu(H))$ iff $uv \in E(H)$ iff $u^0v^0 \in E(\mu(H))$; thus $|K| \leq \omega(H)$. Conversely, every clique of H lifts to V^0 , and $\{r, v^1\}$ is a clique of size 2. Hence $\omega(\mu(H)) = \max\{\omega(H), 2\}$. $\square$

Proposition 3.1 (The seed). *The graph $G = \overline{C_{13}(1, 5)}$ satisfies*

$$\omega(G) = 4, \quad \alpha(G) = 2, \quad \chi(G) = 7, \quad \zeta(G) = 4,$$

and the partition

$$A = \{0, 2, 4, 6\}, \quad B = \{1, 5, 7, 11\}, \quad C = \{3, 10, 12\}, \quad I = \{8, 9\}$$

of $\mathbb{Z}_{13}$ is a cochromatic partition of G into four parts in which A, B, C are cliques and I is independent. A proper 7-colouring of G is given by the classes $\{0, 1\}, \{2, 3\}, \{4, 5\}, \{6, 7\}, \{8, 9\}, \{10, 11\}, \{12\}$.

Proof. Put $D = \{\pm 1, \pm 5\} = \{1, 5, 8, 12\} \subseteq \mathbb{Z}_{13}$, so that $xy \in E(F)$ iff $x - y \in D$.

F is *triangle-free*. The differences of two distinct elements of D are, up to sign, $5 - 1 = 4$, $8 - 1 = 7 \equiv -6$, $12 - 1 = 11 \equiv -2$, $8 - 5 = 3$, $12 - 5 = 7 \equiv -6$, $12 - 8 = 4$; so they lie in $\{\pm 2, \pm 3, \pm 4, \pm 6\}$, which is disjoint from D . Translations are automorphisms of F , so a triangle could be translated to contain 0; its other two vertices would be distinct $a, b \in D$ with $a - b \in D$, which is impossible. Hence $\omega(F) = 2$ (and F has edges).

$\alpha(F) = 4$. The set $\{0, 2, 4, 6\}$ has pairwise cyclic distances 2, 4, 6, none equal to 1 or 5, so it is independent and $\alpha(F) \geq 4$. Suppose F had an independent set of five vertices. Listing them in cyclic order around $\mathbb{Z}_{13}$ gives five cyclic gaps $g_1, \dots, g_5 \geq 1$ with $\sum g_i = 13$, and $g_i \geq 2$ since vertices at cyclic distance 1 are adjacent. The multisets of five integers ≥ 2 with sum 13 are exactly $\{5, 2, 2, 2, 2\}$, $\{4, 3, 2, 2, 2\}$ and $\{3, 3, 3, 2, 2\}$. Two cyclically consecutive gaps summing to 5 mean two chosen vertices at cyclic distance 5, which are adjacent; a single gap 5 likewise. In the first pattern there is a gap of 5. In the second, the unique gap 3 has two cyclic neighbours among $\{4, 2, 2, 2\}$, at most one of which is the 4, so it is adjacent to a 2. In the third, if no 3 were adjacent to a 2, the two 2's and the three 3's would each form a union of arcs of the 5-cycle of gaps with no edge between the two classes, which is impossible in a connected cycle. In every case we obtain two adjacent chosen vertices, a contradiction. Hence $\alpha(F) = 4$.

Parameters of G . Complementation exchanges cliques and independent sets, so $\omega(G) = \alpha(F) = 4$ and $\alpha(G) = \omega(F) = 2$.

$\chi(G) = 7$. Every colour class of a proper colouring of G is independent, hence of size at most 2, so $\chi(G) \geq \lceil 13/2 \rceil = 7$. Each of $\{0, 1\}$, $\{2, 3\}$, $\{4, 5\}$, $\{6, 7\}$, $\{8, 9\}$, $\{10, 11\}$ is a pair at cyclic distance 1, i.e. an edge of F and a non-edge of G , and $\{12\}$ is a singleton; so the displayed classes form a proper 7-colouring of G , and $\chi(G) = 7$.

$\zeta(G) = 4$. Every homogeneous set of G has size at most $\max\{\omega(G), \alpha(G)\} = 4$, so $\zeta(G) \leq \lceil 13/4 \rceil = 4$. The sets A, B, C, I partition $\mathbb{Z}_{13}$. Inside $A = \{0, 2, 4, 6\}$ the cyclic distances are 2, 4, 6, 2, 4, 2; inside $B = \{1, 5, 7, 11\}$ they are 4, 6, 3, 2, 6, 4 (from $5 - 1 = 4$, $7 - 1 = 6$, $11 - 1 \equiv -3$, $7 - 5 = 2$, $11 - 5 = 6$, $11 - 7 = 4$); inside $C = \{3, 10, 12\}$ they are 6, 4, 2 (from $10 - 3 \equiv -6$, $12 - 3 \equiv -4$, $12 - 10 = 2$). None equals 1 or 5, so A, B, C are independent in F , i.e. cliques in G . The vertices 8, 9 are at cyclic distance 1, adjacent in F , so $I = \{8, 9\}$ is independent in G . Thus A, B, C, I is a cochromatic partition of G into four parts with an independent part, and $\zeta(G) = 4$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02/03) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/2408.02400__00/`. Treat it as machine output, not as an authority.

Target writeup: `attacks/2408.02400__00/output.md` Problem: arXiv:2408.02400 (R. Steiner), Problem 1.5 — for every $k \geq 5$, is there a graph G_k with $\omega(G_k) < 5$, $\zeta(G_k) = k$ and $\chi(G_k) = \zeta(G_k) + 3$?

B.1 Interpretation

The writeup's formalization matches the source paper exactly. I fetched arXiv:2408.02400v2 (HTML full text); Problem 1.5 there reads verbatim: "Let $k \geq 5$. Is there a graph G_k with $\omega(G_k) < 5$, $\zeta(G_k) = k$ and $\chi(G_k) = \zeta(G_k) + 3$?" The writeup produces, for every $k \geq 5$, an explicit graph with $\omega = 4$ (< 5), $\zeta = k$ exactly, and $\chi = k + 3$ exactly — precisely the intended reading, with no weakening (it does not settle for $\chi \geq k + 3$ or $\zeta \geq k$; both are pinned exactly by matching upper and lower bounds). There is no interpretation gaming: a reasonable author of the source paper would consider Problem 1.5 answered affirmatively by this construction. Context check: Steiner's Theorem 1.4 (also confirmed against the paper) already gives infinitely many graphs with $\omega < 5$, $\zeta = 4$,

$\chi = 7$ — i.e., gap 3 at $\zeta = 4$ — and Problem 1.5 asks to push ζ to every $k \geq 5$ while keeping the gap at 3; that is exactly what the writeup does. (Note: the catalog page’s paraphrase of Theorem 1.4 — “gap at least 2 ... arbitrarily large ζ ” — is inaccurate relative to the paper, but the Problem 1.5 statement in prompt.md is quoted verbatim and is what the writeup solves, so this does not affect the audit.)

The writeup’s verdict block says “proved”, meaning the yes/no question is answered YES with proof; that is the correct semantics.

B.2 Step-by-step findings

step	label	note
1. Mycielskian setup (Sec. 1)	VALID	Standard definition of $\mu(H)$: old layer V^0 inducing H , independent first layer V^1 with $u^0 v^1$ edges for $uv \in E(H)$, root r joined to all of V^1 . The stated consequences (V^1 independent, r anticomplete to V^0 , $v^0 v^1$ never adjacent) are immediate.
2. Lemma 1, lower bound, clique case	VALID	If the part C containing r is a clique, then C is a subset of $\{r\}$ plus V^1 (r ’s neighborhood), and since V^1 is independent, C contains at most one first-layer vertex and no old vertex. So the remaining $q-1$ parts cover all of V^0 , which induces a copy of H , and restrictions of cliques/independent sets to an induced subgraph stay homogeneous. Hence $\zeta(H) \leq q-1$. Checked in full detail.

step	label	note
3. Lemma 1, lower bound, independent case	VALID	If C is independent, $C \cap V^1$ is empty (r dominates V^1), so $C = \{r\} \cup S^0$ with S independent in H. The sets $D^* = \{v : v^0 \text{ in } D\} \cup \{v \text{ in } S : v^1 \text{ in } D\}$ partition $V(H)$: a vertex v not in S is picked up exactly once via v^0 (which lies in a unique $D \neq C$), a vertex v in S exactly once via v^1 (v^0 lies in C; v^1 lies in a unique $D \neq C$). Homogeneity: for independent D, old-old pairs are nonadjacent in H; an old-new pair u, v (v in S, v^1 in D, u^0 in D) has $u \neq v$ (else v^0 in D, but v^0 in C) and uv not an edge ($v^1 u^0$ would be an edge of $\mu(H)$ inside independent D); new-new pairs are nonadjacent since S is independent. For a clique D (which cannot contain r, and contains at most one first-layer vertex since V^1 is independent), replacing that v^1 by v preserves the clique: $v^1 u^0$ in $E(\mu(H))$ iff uv in $E(H)$, and v^0 not in D (v in S forces v^0 in C). At most $q-1$ nonempty parts, so $\zeta(H) \leq q-1$. Both cases give $\zeta(\mu(H)) \geq \zeta(H)+1$. I re-derived every subcase; no gap. Also tested exhaustively by computer (see below).

step	label	note
4. Lemma 1, equality and persistence	VALID	Given a minimum cochromatic partition $I, P_2, \dots, P_{\zeta}$ of H with I independent, the partition $\{r\} \cup I^0, P_2^0, \dots, P_{\zeta}^0, V^1$ of $\mu(H)$ is cochromatic with $\zeta(H)+1$ parts: $\{r\} \cup I^0$ is independent (r anticomplete to V^0), each P_i^0 inherits its type from H, V^1 is independent. Combined with step 3's lower bound this gives equality, and both $\{r\} \cup I^0$ and V^1 are independent parts, so the hypothesis persists for the next iteration. VALID.
5. Standard Mycielski identities	VALID	$\chi(\mu(H)) = \chi(H)+1$ is Mycielski's classical theorem (1955); the writeup's self-contained lower-bound argument (recolor v by the color of v^1 when v^0 wears r's color) is the standard correct proof — I checked the three edge cases (both endpoints old-colored; one recolored; both recolored is impossible since adjacent v^0, u^0 cannot share r's color). $\omega(\mu(H)) = \max(\omega(H), 2)$: a clique containing r lies in $\{r\} \cup V^1$ hence has size ≤ 2; a clique avoiding r has at most one first-layer vertex v^1 and cannot also contain v^0 (nonadjacent), so projecting v^1 to v is injective and yields a clique of H. $\{r, v^1\}$ realizes 2. VALID (and computationally confirmed: ω stays 4 through two iterations).

step	label	note
6. Seed: F triangle-free (Sec. 2.1)	VALID	$D = \{+1, +5\}$ in $Z_{13} = \{1, 5, 8, 12\}$; pairwise differences of distinct elements are $\{+-2, +-3, +-4, +-6\}$ (re-derived: $5-1=4$, $8-1=7=-6$, $12-1=11=-2$, $8-5=3$, $12-5=7=-6$... all in $\{2,3,4,6,7,9,10,11\} = +- \{2,3,4,6\}$), disjoint from D, and translation-invariance reduces any triangle to 0, a, b with a, b, a-b in D. Confirmed by brute force: $\omega(F) = 2$.
7. Seed: $\alpha(F) = 4$ (Sec. 2.2)	VALID	$\{0,2,4,6\}$ has pairwise cyclic distances $\{2,4,6\}$, avoiding $\{1,5\}$: independent. For the upper bound, 5 vertices give 5 cyclic gaps, each ≥ 2, summing to 13, so the multiset is $(5,2,2,2,2)$, $(4,3,2,2,2)$ or $(3,3,3,2,2)$; a gap of 5 is an edge, and in the last two cases a 3-gap must be cyclically adjacent to a 2-gap (counting argument re-checked: the single 3 has two neighbors among $\{4,2,2,2\}$, only one can be the 4; three 3's and two 2's in a 5-cycle force a 3-2 adjacency), giving two chosen vertices at distance 5. Each case yields a contradiction, so $\alpha(F) = 4$. (The argument does not need to consider longer gap-sums such as three consecutive gaps summing to $8 = 13-5$, because every case is already killed by pairs.) Confirmed by brute force: $\alpha(F) = 4$.
8. $\omega(G) = 4$, $\alpha(G) = 2$	VALID	Complementation swaps clique and independence numbers. Brute force: $\omega(G) = 4$, $\alpha(G) = 2$.
9. $\chi(G) = 7$ (Sec. 2.3)	VALID	Lower bound $13/\alpha(G) = 13/2$ gives $\chi \geq 7$. The explicit classes $\{0,1\}$, $\{2,3\}$, ..., $\{10,11\}$, $\{12\}$ are pairs at circulant distance 1, i.e., edges of F, i.e., nonedges of G. Both verified by computer: exact $\chi(G) = 7$ and the explicit 7-coloring is proper.

step	label	note
10. $\text{zeta}(G) = 4$ with independent part (Sec. 2.4)	VALID	Every homogeneous set of G has size $\leq \max(\omega, \alpha) = 4$, so $\text{zeta} \geq \lceil 13/4 \rceil = 4$. The partition $A = \{0,2,4,6\}$, $B = \{1,5,7,11\}$, $C = \{3,10,12\}$, $I = \{8,9\}$ is checked: internal cyclic distances of A, B, C avoid $\{1,5\}$ (re-derived pairwise: e.g. B has distances 4, 6, 3, 2, 6, 4), so they are cliques of G ; 8-9 is an F-edge, so I is independent in G . Exact computation confirms $\text{zeta}(G) = 4$ and this partition. $G_k = \mu^{\{k-4\}}(G)$. Induction hypothesis (a minimum cochromatic partition with an independent part) holds for the seed (step 10) and persists (step 4), so zeta increments by exactly 1 each step: $\text{zeta}(G_k) = k$. $\chi(G_k) = 7 + (k-4) = k+3$ and $\omega(G_k) = 4$ by step 5. Order formula $14 \cdot 2^{k-4} - 1$ satisfies the recurrence $n \rightarrow 2n+1$ from 13. All exact.
11. Iteration (Sec. 3)	VALID	

No step is a GAP or ERROR.

B.3 Reference check

- **arXiv:2408.02400 (Steiner, v2)** — fetched (abstract page and HTML full text). Problem 1.5 exists and reads exactly as in the prompt and the writeup: “Let $k \geq 5$. Is there a graph G_k with $\omega(G_k) < 5$, $\text{zeta}(G_k) = k$ and $\chi(G_k) = \text{zeta}(G_k)+3$?” Theorem 1.4 reads: “There are infinitely many graphs G with $\omega(G) < 5$, $\text{zeta}(G) = 4$ and $\chi(G) = 7$ ” (proved there via an 11-vertex gadget of two joined C_5 ’s, not via circulants). Conjecture 1.2 (Erdos-Gimbel-Straight 1988): every G with $\omega(G) < 5$ and $\text{zeta}(G) > 3$ satisfies $\chi(G) \leq \text{zeta}(G)+2$ — which Steiner disproves; Problem 1.5 asks for gap exactly 3 at every $\text{zeta} = k \geq 5$.
- **Mycielski identities** — $\chi(\mu(H)) = \chi(H)+1$ and preservation of clique number are textbook (Mycielski 1955); the writeup in any case re-proves the parts it needs, so nothing is taken on faith.
- **Priority / novelty** — Semantic Scholar lists exactly two papers citing 2408.02400: arXiv:2408.13839 (Heckel) and arXiv:2409.17614, both on the Erdos-Gimbel random-graph question, not Problem 1.5. Additional web searches (including an August 2026 random-graph paper, arXiv:2608.30604, which explicitly does not touch deterministic bounded-clique constructions) found no resolution of Problem 1.5, consistent with the catalog’s May 2026 review. The writeup does not cite anything unverifiable.

B.4 Computational check

Scripts in `verification/scripts/2408.02400__00/` (pure Python 3, no external graph libraries needed): `graphlib.py` (Bron-Kerbosch, exact chromatic number, exact cochromatic number via branch-and-bound set cover over maximal cliques and maximal independent sets), `check_seed.py`, `check_lemma.py`, `check_iterate.py`, `check_iterate2.py`.

Results obtained:

1. **Seed** (`check_seed.py`): for $F = C_{13}(1,5)$ and G its complement — $\omega(F) = 2$, $\alpha(F) = 4$, $\omega(G) = 4$, $\alpha(G) = 2$, exact $\chi(G) = 7$, exact $\zeta(G) = 4$; the explicit 7-coloring and the explicit cochromatic partition (A, B, C cliques, $I = \{8,9\}$ independent) are valid. All claimed seed values are exactly right.
2. **Lemma 1** (`check_lemma.py`): for every one of the 1099 labeled graphs on 1-5 vertices, and for 95 random graphs on 6-8 vertices at various densities, both assertions hold: $\zeta(\mu(H)) \geq \zeta(H)+1$ always, and when H has a minimum cochromatic partition with an independent part, $\zeta(\mu(H)) = \zeta(H)+1$ with the property persisting. No counterexample.
3. **First iterate, $k = 5$** (`check_iterate.py`): $\mu(G)$ on 27 vertices computed directly — $\omega = 4$, exact $\zeta = 5$, exact $\chi = 8$, and an optimal cochromatic partition with an independent part exists. This independently verifies the $k = 5$ instance $(\omega, \zeta, \chi) = (4, 5, 8) = (4, k, k+3)$ without using Lemma 1.
4. **Second iterate, $k = 6$** (`check_iterate2.py`): $\mu^2(G)$ on 55 vertices — $\omega(\mu^2(G)) = 4$ confirmed by direct computation (Bron-Kerbosch on the 55-vertex graph). The exact ζ/χ computation at this size was aborted after about 15 minutes of branch-and-bound (beyond the compute budget), so the exact values for $k \geq 6$ rest on Lemma 1 and the Mycielski chromatic identity, which are proven in full and machine-tested as above.

No claimed property failed; every number the writeup states that was within reach of brute force came out exactly as claimed.

B.5 Caveats

- For $k \geq 6$ the parameters are established by the (fully verified) induction, not by direct computation; this is a normal state of affairs for an infinite family, and the base case and first inductive step were both verified independently by machine.
- The writeup’s remark that the construction “also gives such a graph for $k = 4$ ” is true (the seed itself) but not new — it is an instance of Steiner’s Theorem 1.4.
- The lemma’s equality direction needs the independent-part hypothesis; the writeup states this and discharges it at every step (seed partition has $I = \{8,9\}$; persistence proven). Without it only $\zeta(\mu(H)) \geq \zeta(H)+1$ would follow, which would still give $\zeta(G_k) \geq k$ but not equality; no such gap arises here.
- Edge/convention checks: the Mycielskian convention used (root adjacent to the shadow layer only) is the standard one and is self-contained in the writeup; the identity $\omega(\mu(H)) = \max(\omega(H), 2)$ is applied only with $\omega = 4$, where it is safe; $k \geq 5$ is the range demanded by Problem 1.5 and is fully covered.
- The writeup’s own caveat is only bibliographic priority; my search found no prior resolution of Problem 1.5, so the result appears to be new, though a definitive novelty claim would require the usual human literature diligence.

B.6 Referee summary

I set out to break this proof and could not. The argument has exactly two components, both of which survive maximal scrutiny. (1) The 13-vertex seed $G = \text{complement of } C_{13}(1,5)$: every claimed parameter ($\omega = 4$, $\alpha = 2$, $\chi = 7$, $\zeta = 4$, and the existence of an optimal cochromatic partition

with an independent part) was verified by exhaustive computation, and the hand proofs of each are correct as written. (2) Lemma 1, a genuinely nice observation that the Mycielskian increases the chromatic number by at least one, with equality when an optimal partition has an independent part and with that property self-perpetuating: I re-derived every subcase of the partition-transfer argument (including the subtle points — the part containing the root meets at most one shadow vertex when it is a clique and no shadow vertex when it is independent; v in S forces v^0 into C so replacements never collide) and found it airtight, and machine-tested it exhaustively on all graphs up to 5 vertices. The $k = 5$ instance was additionally verified end-to-end by brute force on the 27-vertex Mycielskian. The interpretation matches Steiner’s Problem 1.5 verbatim, the references check out, and no follow-up literature resolves the problem. This is a correct, complete, and apparently novel affirmative solution of Problem 1.5.

References

- [1] J. Mycielski, Sur le coloriage des graphes, *Colloq. Math.* 3 (1955) 161–162.
- [2] R. Steiner, On the difference between the chromatic and cochromatic number, arXiv:2408.02400 (2024; v2).